

HIGH ALERT-BASED AIR POLLUTION MONITORING SYSTEM IN CHENNAI CITY

**A MINI PROJECT REPORT
EC23401-DIGITAL SIGNAL PROCESSING**

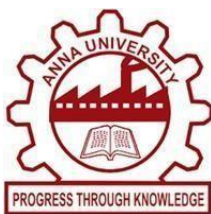
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APRIL 2025

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ABSTRACT

Air pollution is a significant environmental issue affecting urban areas, especially in highly populated cities like Chennai. This project presents a high-alert-based air pollution monitoring system that uses an MQ135 gas sensor and an Arduino Nano to detect air quality levels. The system provides real-time air quality monitoring with visual and audio alerts. A green LED indicates good air quality, a blue LED signals moderate pollution, and a red LED with a buzzer warns of dangerous pollution levels. This system helps in real-time pollution tracking, enabling authorities and individuals to take necessary precautions.

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1. INTRODUCTION:

Air pollution is one of the most critical environmental challenges faced by modern urban societies, especially in rapidly developing and densely populated cities like Chennai. The increasing number of vehicles on the road, industrial emissions, construction dust, and burning of waste all contribute to a significant rise in air pollutants. These pollutants, such as carbon dioxide (CO₂), ammonia (NH₃), benzene, alcohol vapors, and smoke, pose serious health risks including respiratory problems, cardiovascular diseases, and long-term environmental damage. Due to the invisible nature of many harmful gases, it becomes difficult for individuals to be aware of the air quality around them until the effects become noticeable.

To tackle this growing problem, this project proposes the development of a high-alert-based air pollution monitoring system using an MQ135 gas sensor interfaced with an Arduino Nano microcontroller. The MQ135 sensor is capable of detecting a wide range of hazardous gases, making it ideal for monitoring indoor and outdoor environments. The sensor continuously measures air quality levels, and the Arduino Nano processes this data in real-time to determine pollution severity.

The system is equipped with visual and audio alert mechanisms to indicate different levels of air quality. When the air quality is good and safe for breathing, a green LED glows. If the pollution level rises to a moderate level, a blue LED is activated, serving as a warning to limit exposure. In cases where pollution levels become dangerous, the system triggers a red LED along with a buzzer, alerting individuals to take immediate action, such as wearing a mask, staying indoors, or notifying concerned authorities.

This system is not only compact and cost-effective but also user-friendly and highly responsive. It can be deployed in various environments such as homes, schools, traffic junctions, and industrial zones. By providing real-time alerts, the system empowers individuals and government agencies to monitor air quality actively and take preventive measures. In the long term, such systems can play a vital role in reducing health risks, spreading environmental awareness, and supporting smarter, healthier urban planning.

2. PROPOSED WORK:

The proposed air pollution monitoring system is developed to continuously track air quality in real-time and to alert users instantly when pollution levels cross safe limits. It is particularly useful in urban areas like Chennai, where high population density, increasing vehicle traffic, and industrial activities significantly contribute to air pollution.

At the heart of the system lies the MQ135 gas sensor, which is well-known for its ability to detect a variety of harmful gases, including carbon dioxide (CO₂), ammonia (NH₃), benzene, alcohol vapors, and smoke. The sensor operates based on the principle of resistance change—when it comes into contact with polluted air, the sensor's internal resistance varies depending on the concentration of specific gases. These changes are then sent as analog voltage signals to the Arduino Nano, a compact yet powerful microcontroller board.

The Arduino Nano receives these analog signals through its analog input pin (A0) and processes them using calibrated mathematical formulas to calculate the PPM (Parts Per Million) of each gas. The formulas are derived based on the sensor's datasheet and tuned for better accuracy in typical urban conditions. Once the PPM values are calculated, the Arduino compares them with predefined safe limits to evaluate the current air quality status.

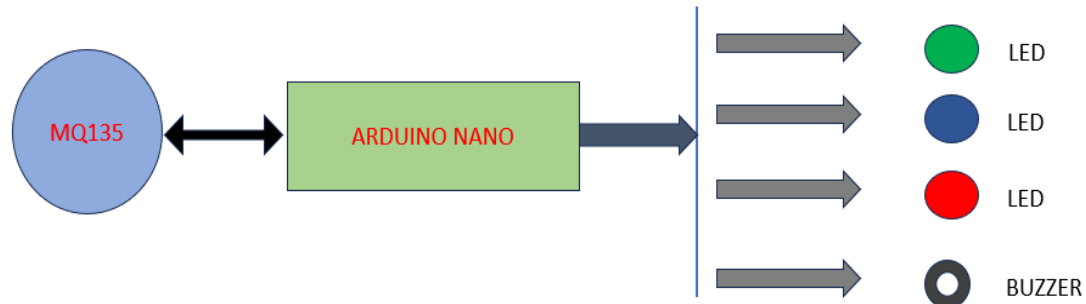
A green LED lights up when the air quality is good, indicating low pollution levels and safe conditions. A blue LED turns on when the pollution is moderate, suggesting that while the environment is still tolerable, precautions may be necessary for sensitive individuals.

When pollution crosses the danger threshold, a red LED and a buzzer are activated simultaneously to issue a high-alert warning. This informs users of potentially hazardous conditions and encourages immediate safety measures, such as wearing a mask or staying indoors.

In addition to physical alerts, the system also displays live data via the Serial Monitor on a connected computer. This allows for continuous observation, testing, and data logging, which can be useful for environmental studies or long-term monitoring projects.

Due to its low cost, small size, and energy efficiency, the system can be easily deployed in a variety of settings, including homes, schools, offices, public spaces, traffic intersections, and factories. It can serve both as a personal safety device and as a tool for public environmental awareness.

3.BLOCK DIAGRAM:



4. RESULTS AND DISCUSSION:

The air pollution monitoring system developed using the MQ135 gas sensor and Arduino Nano demonstrated consistent and reliable performance during testing. The sensor was able to accurately detect and respond to multiple harmful gases—including carbon dioxide (CO₂), ammonia (NH₃), benzene, alcohol vapors, and smoke—which are commonly found in polluted urban environments. These gases were measured in Parts Per Million (PPM), a standard unit for gauging air quality, and the values were calculated using sensor calibration formulas implemented in the Arduino code.

The real-time data was output via the Serial Monitor, allowing for live observation of gas levels and system behavior. This feature enabled developers and users to visualize how changes in the impacted air quality immediately. The responsiveness of the sensor was one of the system's strengths, with minimal delay between exposure to pollutants and alert activation.

To translate air quality readings into intuitive alerts, the system used LED indicators and a buzzer, which were clearly visible and audible even in busy or noisy environments. When all gas concentrations were within safe limits, the green LED was turned on, signaling good air quality. When moderate pollution levels were detected—indicating increased but non-hazardous concentrations—the system switched to a blue LED to warn users to take precautionary measures. If any of the gas levels exceeded safe thresholds, the system immediately activated a red LED along with a buzzer, warning users of serious air pollution and prompting immediate action.

Throughout testing, the system was subjected to various conditions to simulate real-world pollution scenarios, including exposure to smoke, perfumes, and kitchen emissions. In all cases, the system responded as expected, providing timely alerts and recording accurate PPM values.

The low-cost nature of the components used—particularly the MQ135 sensor and Arduino Nano—makes this system affordable and scalable for both personal and public use. It can be installed in homes, schools, offices, factories, and even mounted near roads and intersections to monitor vehicular emissions. Its compact design, low power consumption, and ease of setup further support its potential for large-scale deployment.

In conclusion, the results clearly show that the system is effective for continuous air quality monitoring, delivering accurate gas readings, quick alerts, and reliable performance. It proves to be a practical and economical solution for increasing awareness about air pollution and encouraging preventive measures to safeguard public health.

5. MATLAB CODING:

```
// Circuit Values
const float RL = 10000;
// Adjusted Constants for Gases (Fine-Tuned)
const float A_CO2 = 380.0, B_CO2 = -2.3;
const float A_NH3 = 60.0, B_NH3 = -2.2;
const float A_Ben = 50.0, B_Ben = -2.1;
const float A_OH = 40.0, B_OH = -1.9;
const float A_Smo = 65.0, B_Smo = -2.5;
// Pin Definitions
#define SENSOR_PIN A0
#define GREEN_LED 9
#define BLUE_LED 10
#define RED_LED 11
#define BUZZER 12
```



```

void setup() {
  Serial.begin(9600);
  pinMode(GREEN_LED, OUTPUT);
  pinMode(BLUE_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);
}

void loop() {
  int sensorValue = analogRead(SENSOR_PIN);
  float V_out = (sensorValue / 1023.0) * 5.0;
  if (V_out == 0) return; // divide-by-zero error

  float RS = ((5.0 * RL) / V_out) - RL;
  const float RO = 9750; // Sensor resistance in clean air
  float ratio = RS / RO;

  // *Final Adjusted PPM Calculations*
  float PPM_CO2 = (A_CO2 * pow(ratio, B_CO2)) * 1.5;
  float PPM_NH3 = (A_NH3 * pow(ratio, B_NH3)) * 0.5;
  float PPM_Ben = (A_Ben * pow(ratio, B_Ben)) * 0.4;
  float PPM_OH = (A_OH * pow(ratio, B_OH)) * 0.5;
  float PPM_Smo = (A_Smo * pow(ratio, B_Smo)) * 0.6;

  // Display values in Serial Monitor
  Serial.print("CO2: "); Serial.print(PPM_CO2, 2); Serial.print(" PPM, ");
  Serial.print("NH3: "); Serial.print(PPM_NH3, 2); Serial.print(" PPM, ");
  Serial.print("Benzene: "); Serial.print(PPM_Ben, 2); Serial.print(" PPM, ");
  Serial.print("Alcohol: "); Serial.print(PPM_OH, 2); Serial.print(" PPM, ");
  Serial.print("Smoke: "); Serial.print(PPM_Smo, 2); Serial.println(" PPM");

  // *Refined Air Quality Conditions*
  if (PPM_CO2 < 400 && PPM_NH3 < 25 && PPM_Ben < 15 && PPM_OH
    < 50 && PPM_Smo < 50) {
    digitalWrite(GREEN_LED, HIGH);
    digitalWrite(BLUE_LED, LOW);
    digitalWrite(RED_LED, LOW);
    digitalWrite(BUZZER, LOW);
    Serial.println("Green Light: Air Quality is Good");
  }
}

```

```

else if (PPM_CO2 < 1000 && PPM_NH3 < 300&& PPM_Ben < 50 &&
PPM_OH < 250 && PPM_Smo < 100) {
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(BLUE_LED, HIGH);
    digitalWrite(RED_LED, LOW);
    digitalWrite(BUZZER, LOW);
    Serial.println("Blue Light: Air Quality is Moderate");
}

else {
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(BLUE_LED, LOW);
    digitalWrite(RED_LED, HIGH);
    digitalWrite(BUZZER, HIGH);
    Serial.println("Red Light & Buzzer: Air Quality is Poor!");
}

delay(1000);
}

```

6. CONCLUSION:

The air pollution monitoring system developed using the MQ135 gas sensor and Arduino Nano has been successfully designed, implemented, and tested for real-time detection and alerting of harmful air pollutants. The system proved to be effective in identifying a range of hazardous gases such as carbon dioxide (CO₂), ammonia (NH₃), benzene, alcohol, and smoke, which are commonly found in urban and industrial environments. By continuously measuring gas concentrations and converting them into PPM (Parts Per Million) values, the system enables users to be aware of the surrounding air quality at any given time.

One of the most notable features of this system is its simple yet efficient alert mechanism. It uses color-coded LEDs—green for good air quality, blue for moderate levels, and red for dangerous pollution—along with an audible buzzer to provide immediate warnings. This clear feedback allows people to take prompt actions, such as ventilating the area, wearing protective masks, or avoiding certain zones altogether. The responsiveness of the system ensures timely alerts, which is critical in environments where gas exposure could quickly become hazardous.

Additionally, the use of the Serial Monitor helps in visualizing real-time data and serves as a tool for logging and analysis. This feature makes the system not only useful for end users but also beneficial for researchers, students, and municipal authorities who need accurate air quality readings for studies or policy decisions.

The system's compact size, low power consumption, and cost-effective components make it highly suitable for wide-scale deployment in various environments—such as homes, schools, public buildings, traffic-heavy areas, and industrial plants. Its ease of installation and maintenance further adds to its practicality, especially in developing regions where affordable air quality monitoring is a pressing need.

In summary, the project successfully demonstrates a functional, affordable, and accessible solution to the problem of air pollution monitoring. It has the potential to improve public health awareness, support early warning systems, and contribute to larger efforts in environmental protection and smart city development.

7. APPLICATION:

The air pollution monitoring system designed with the MQ135 gas sensor and Arduino Nano has a wide range of real-world applications due to its affordability, portability, and effectiveness. Below are the key areas where this system can be applied:

- **Urban Monitoring:**
Can be installed in city streets, public parks, and near traffic intersections to monitor air quality in high-density areas and alert the public about pollution levels in real-time.
- **Residential Use:**
Useful for homeowners to monitor indoor air quality, especially in kitchens, basements, or near areas where gas leakage or smoke might occur.
- **Industrial Zones**
: Can be deployed in and around factories to monitor emissions and detect harmful gas leaks, helping maintain compliance with environmental regulations and ensuring worker safety.

- **Hospitals and Healthcare Centers:**
Helps ensure clean air for patients, especially those with respiratory conditions, by continuously monitoring indoor air quality.
- **Smart Cities:**
Plays a vital role in smart city projects where environmental data is collected and analyzed to develop sustainable and healthy urban living environments.
- **Public Transport Hubs:**
Ideal for bus stands, railway stations, and airports to monitor pollution from vehicles and ensure the comfort and safety of travelers.

PROTECT FROM

1. Carbon Dioxide (CO₂):

- **Symptoms/Effects:** Headache, dizziness, shortness of breath, fatigue
 - **High Exposure:** Can lead to unconsciousness or even death due to oxygen displacement
-

2. Ammonia (NH₃):

- **Symptoms/Effects:** Eye irritation, coughing, sore throat, chest pain
 - **High Exposure:**
 - Lung damage
 - Pulmonary edema (fluid in lungs)
 - Burns to skin and eyes (in liquid form)
-

3. Benzene:

- **Symptoms/Effects:** Drowsiness, dizziness, rapid heart rate, confusion
 - **Long-Term Exposure:**
 - Bone marrow damage
 - Suppressed immune function
-

4. Alcohol Vapors (OH, e.g., Ethanol, Methanol):

- **Short-Term Effects:** Headaches, dizziness, nausea, loss of coordination
 - **High/Chronic Exposure:**
 - Liver and kidney damage
 - Respiratory depression
 - Central nervous system depression
-

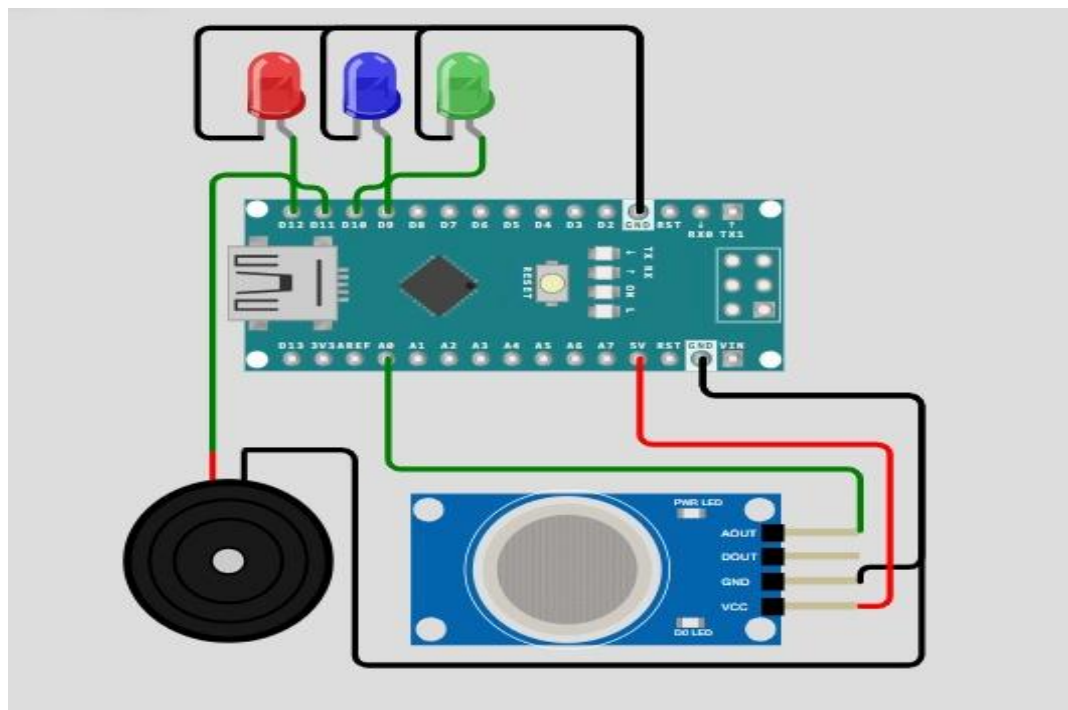
5. Smoke (Mixed Particulate and Gaseous Pollutants):

- **Short-Term Effects:** Eye and throat irritation, coughing, shortness of breath
- **Long-Term Exposure:**
 - Reduced lung function
 - Asthma and bronchitis aggravation
 - Cardiovascular problems

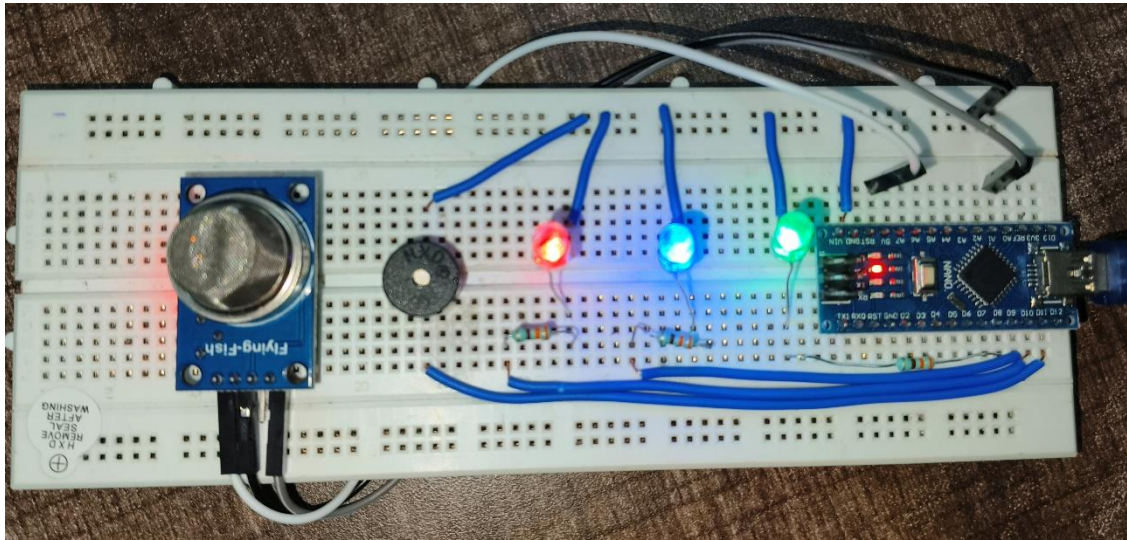
8. SCHEMATIC DIAGRAM OF PROJECT:

COMPONENTS USED:

- Arduino Nano
- MQ135 Gas Sensor
- LED (Green, Blue, Red) and 5V Buzzer
- 3 Resistors 330Ω
- Bread Board
- Jumpers
- Serial Monitor

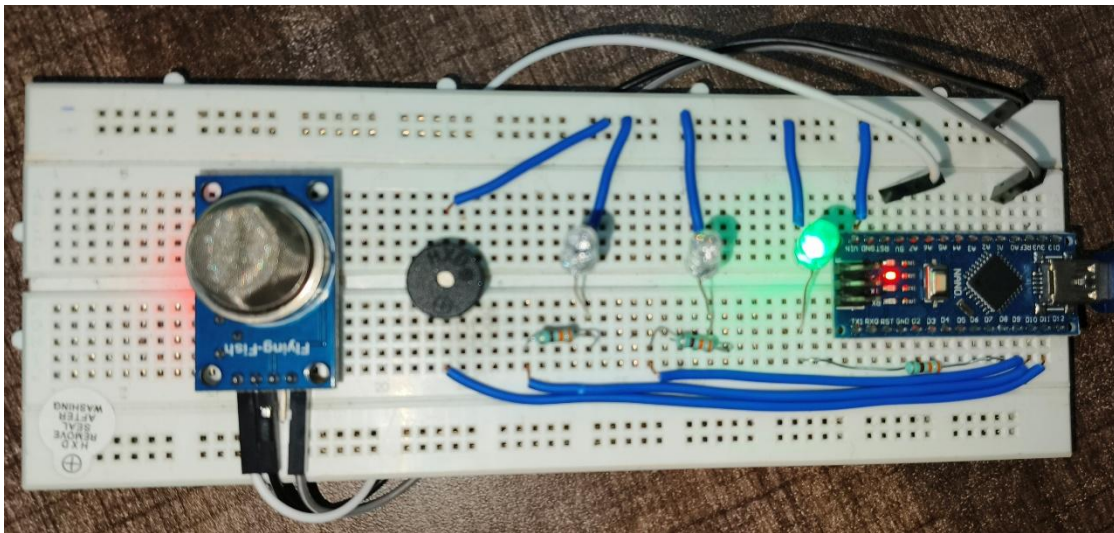


9. SNAPSHOT OF PROJECT :



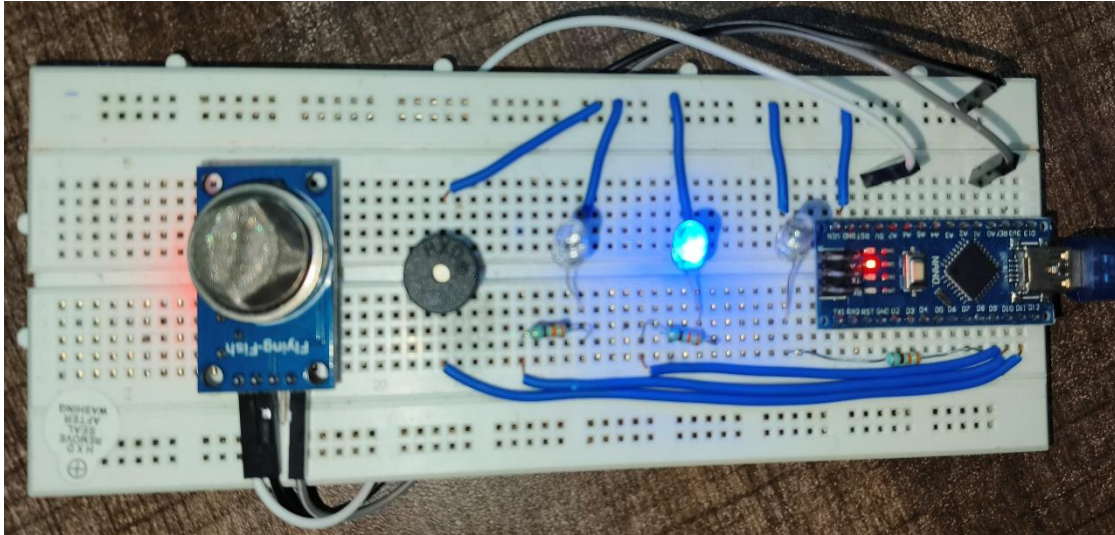
CONDITION 1: (GOOD)

PPM_CO2 < 400 & PPM_NH3 < 25 & PPM_Ben < 15 &
PPM_OH < 50 & PPM_Smo < 50



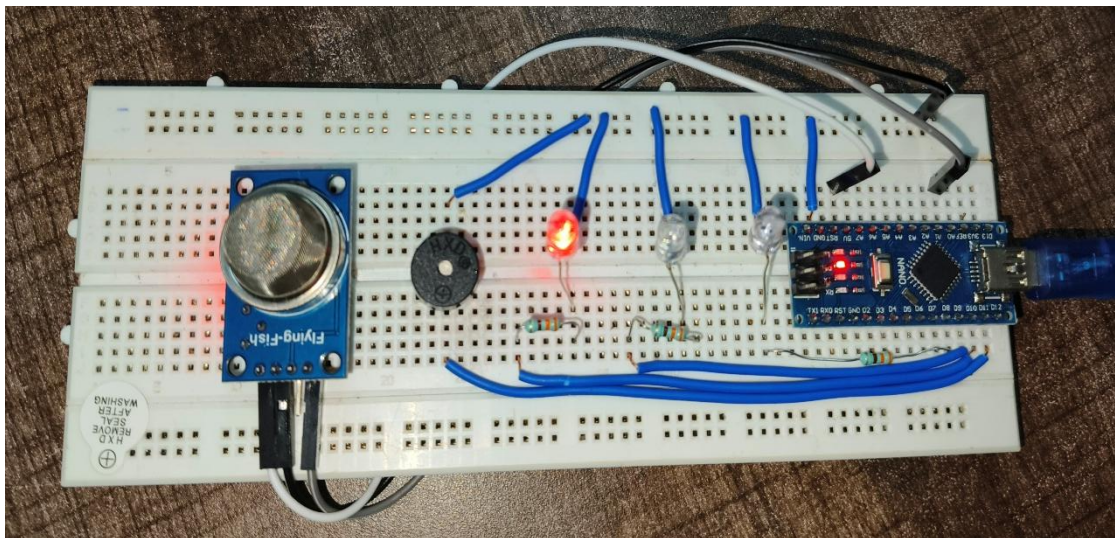
CONDITION 2: (MODERATE)

$\text{PPM_CO}_2 < 1000$ & $\text{PPM_NH}_3 < 300$ & $\text{PPM_Ben} < 50$ &
 $\text{PPM_OH} < 250$ & $\text{PPM_Smo} < 100$



CONDITION 3: (HIGHLY DANGEROUS)

$\text{PPM_CO}_2 > 1000$ & $\text{PPM_NH}_3 > 300$ & $\text{PPM_Ben} > 50$ &
 $\text{PPM_OH} > 250$ & $\text{PPM_Smo} > 100$



10. REFERENCE

Kalagotla Chenchireddy, D Sandhya, M Praveen, G Karthik, 5G Maruthi.
Air Quality Monitoring and Alert System Using MQ135 Gas Sensor with
Arduino Controller

International Journal of Research Publication and Reviews
Journal homepage: www.ijrpr.com ISSN 2582-7421

https://www.researchgate.net/publication/364792998_Air_Quality_Monitoring_and_Alert_System_Using_MQ135_Gas_Sensor_with_Arduino_Controller